Neural Field Perceiver

Cross-Patient Speech Decoding from Intra-Operative μECoG
via Reprojection-Supervised Spatial Alignment

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This document is a **technical design document** for the Neural Field Perceiver project. It unifies the theoretical problem framing, cross-disciplinary foundations from 3D computer vision, the full architecture specification, uncertainty estimation, and the experimental plan. Where possible, claims are labeled as **externally established** (published literature with citation), **internally observed** (our experiments, with protocol noted), or **working hypotheses** (plausible but unverified). Not every claim carries a label; the categorization is strongest in Sections 2–5 and should be treated as a useful habit, not a guarantee of completeness. Structural assumptions that could be wrong are marked in red.

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Part I

Problem Framing

1 The Physical Problem

Each patient's micro-electrocorticography (μECoG) array is a **sparse spatial sample** of neural activity over the cortical surface during speech production. The array is placed subdurally over left ventral sensorimotor cortex (vSMC) during surgery, and its position in standardized brain space (MNI-152) is determined post-hoc via electrode reconstruction (BioImage Suite for DBS patients, Brainlab for tumor patients).

1.1 The cortical surface, not free 3D space

The cortex is a folded two-dimensional sheet, not a volume. Let $\mathcal{S}$ denote the cortical surface embedded in $\mathbb{R}^3$. The high-gamma analytic amplitude (HGA, 70–150 Hz envelope) is a function on this surface:

$$a : \mathcal{S} \times \mathbb{R} \to \mathbb{R} \tag{1}$$

For patient i with N_i electrodes at cortical surface positions $\{\mathbf{p}_j^{(i)}\}_{j=1}^{N_i} \subset \mathcal{S}$, we observe:

$$\boxed{d_j^{(i)}(t) = a\big(\mathbf{p}_j^{(i)},\ t\big) + \epsilon_j^{(i)}(t)} \tag{2}$$

where ϵ captures measurement noise, electrode impedance variation, and sub-electrode-scale neural activity not resolved by the $\sim$2 mm contact spacing.

In practice, we approximate cortical surface positions with 3D MNI coordinates, $\mathbf{p}_j^{(i)} \approx (x, y, z)_{\text{MNI}}^{(i,j)}$, which has two consequences:

1. **Euclidean distance in MNI space does not equal geodesic distance on the cortical surface.** Two electrodes 3 mm apart in MNI may sit on opposite banks of a sulcus, making them functionally distant. For μECoG arrays on exposed cortex, the local patch under the array is approximately planar, so within-array Euclidean distances are reasonable. Cross-patient comparisons at larger scales are more affected by folding geometry.
2. **Surface-based coordinates** (e.g., projection to `fsaverage`) would better respect cortical geometry than volumetric MNI coordinates. Whether surface reconstructions exist for our patients depends on the BioImage Suite / Brainlab outputs available.

The decoding problem. Given observations $\{d_j^{(i)}(t)\}$ at approximately known surface positions, infer the phoneme sequence $\mathbf{y} = (y_1, y_2, y_3)$ that produced the spatiotemporal pattern. This is three separate 9-way classifications, one per phoneme position in the non-word token (e.g., /a/, /b/, /e/ for the token /abe/).

The cross-patient problem. Train on patients $\{1, \ldots, K\}$ and decode for a held-out patient $K+1$ whose electrodes sample the cortical surface at *different positions*. The observations are not aligned: electrode j on patient i and electrode j on patient i' correspond to different cortical locations. This problem admits several evaluation regimes of increasing difficulty (Section 18).

Factor	Quantity	Implication
Electrodes per patient	63–201	Sparse sampling of the cortical field
Electrode spacing	∼2 mm	Samples at approximately columnar scale
Array offset across patients	15–25 mm in MNI	Different cortical regions sampled
Array rotation	Variable	Channel indices meaningless across patients
Correspondence uncertainty	Several mm	Coordinate-based alignment is approximate
Cortical folding variability	Patient-specific	Sulcal geometry differs across individuals
Tumor mass effect (subset)	Variable	Macroscopic warping in 6 of 12 patients
Trials per patient	46–178	Small-data regime for supervised learning
Total utterance per patient	∼1 min	Insufficient for single-patient SSL
Phoneme vocabulary	9 classes × 3 positions	Three 9-way per-position classifications

1.2 What makes this hard

2 The Shared Dynamics Hypothesis

Cross-patient decoding is only possible if the cortical field $a(\mathbf{p}, t)$ shares structure across patients. We decompose:

$$\boxed{a^{(i)}(\mathbf{p}, t) = f\big(T_i(\mathbf{p}),\ t;\ \mathbf{y}\big) + r^{(i)}(\mathbf{p}, t)} \tag{3}$$

where:

- $f(\cdot, t; \mathbf{y})$ is the **shared neural field**—the spatiotemporal pattern that phoneme sequence $\mathbf{y}$ produces in canonical cortical space.
- $T_i : \mathcal{S}_i \to \mathcal{S}_{\text{canonical}}$ is a **patient-specific spatial transform** mapping each patient's cortical surface to a canonical reference. MNI normalization is our estimate of T_i, but it is imperfect.
- $r^{(i)}(\mathbf{p}, t)$ is a **patient-specific residual** capturing individual neural organization, signal characteristics, and functional plasticity.

The viability of cross-patient decoding depends on the ratio $\|f\|/\|r^{(i)}\|$ at the spatial resolution accessible to our measurements and alignment procedure.

How the equations relate to the model. Equations (2)–(3) are the physical setup. Section 3 introduces a further idealization (shared observation function on a manifold). The **implemented model** (Section 10) relaxes both idealizations: it conditions explicitly on patient identity and does not enforce a strictly shared observation function. The earlier equations motivate the architecture; they are not claims that the model instantiates them exactly.

2.1 Evidence for shared dynamics

2.1.1 Externally established (published literature)

Per-patient input layers + shared temporal backbone is a successful transfer architecture in related settings. Singh et al. [1] demonstrate group training with N=25 sEEG patients: per-patient Conv1D + shared BiLSTM (2×64), frozen shared backbone + fine-tuned per-patient layers yields PER 0.49 vs. 0.57 single-subject (p<0.001). Willett et al. [2] use per-day affine input layer (256→256) + shared 5-layer GRU (512 hidden) for day-to-day BCI transfer.

Caveat: Singh's data regime is ∼60 min/patient (vs. our ∼1 min); Conv1D hyperparameters are unreported. Willett is N=1 with ∼40 min/day. The architectural pattern transfers; the hyperparameters

do not.

Articulatory somatotopy in vSMC is reproducible within individuals. Metzger et al. [3] show somatotopic organization of speech articulators in one patient with chronic ECoG (253 channels): labial, front tongue, back tongue, and vocalic representations are spatially organized ($p<0.0001$), preserved even after 18 years of anarthria. Bouchard et al. [8] demonstrated reproducible articulatory maps in vSMC across 3 participants.

Caveat: Metzger: $N=1$ chronic implant. Bouchard: $N=3$, standard ECoG. Cross-patient spatial consistency is inferred from anatomical uniformity but not directly measured in a large-scale study.

Articulatory features are a robust encoding basis in vSMC. Wu et al. [4] reconstruct speaker-invariant articulatory features from HD-ECoG over vSMC (PCC 0.75–0.80, $N=3$), including one intra-operative patient (PCC = 0.76). Anumanchipalli et al. [9] reconstructed continuous articulatory trajectories from motor cortex.

MNI coordinates may help cross-patient ECoG transfer. Chen et al. [5] (SwinTW) use MNI coordinates and ROI embeddings across 43 ECoG + 9 sEEG subjects, reporting mean PCC = 0.765 on unseen participants in cross-validation.

Caveat: Standard ECoG has $\sim$10 mm spacing, comparable to inter-subject correspondence uncertainty. Our μECoG spacing ($\sim$2 mm) is finer, meaning coordinates may encode global array position but not resolve within-array relationships.

Motor cortex neural manifold structure is preserved across individuals. Safaie et al. [6] show motor cortex population dynamics during reaching are geometrically similar across monkeys. Gallego et al. [7] show low-dimensional manifold dynamics are preserved within individuals across time despite neural turnover.

Caveat: Safaie: limb motor cortex, non-human primates. Gallego: within-individual. Extension to cross-individual speech motor cortex is plausible but untested.

2.1.2 Internally observed (our experiments)

Cross-patient supervised transfer works at this data scale. Our LOPO pipeline achieves PER = 0.762 on held-out S14 (LP-FT regime, grouped-by-token CV), compared to per-patient PER = 0.737 (same CV). Direct evidence that decodable shared structure exists in f.

Articulatory bottleneck head improves LOPO decoding. Consistent with f being organized along articulatory feature dimensions.

55 experiments on S14 converge to PER 0.750–0.780. See Section 5.4 for interpretation.

SSL features carry partial structure. JEPA + LP-FT achieves PER = 0.797 (vs. 0.854 basic). Supervised features with the same recipe achieve 0.737.

2.1.3 Working hypotheses (plausible but unverified)

The $\sim$0.76 PER convergence partly reflects spatial alignment limitations (not isolated from data scarcity or S14-specific effects). MNI coordinates may improve decoding (depends on registration accuracy and whether alignment is binding). Cross-task data pooling [12] is a high-value, low-risk path to more data.

3 The Neural Manifold Connection

3.1 Two views of the same phenomenon

The cortical field model and the neural manifold hypothesis are dual descriptions. The bridge is the **latent articulatory state** $\mathbf{z}(t) \in \mathbb{R}^k$:

$$\boxed{d_j^{(i)}(t) = g\big(\mathbf{p}_j^{(i)},\ \mathbf{z}(t)\big) + \epsilon_j^{(i)}(t)} \tag{4}$$

where $g : \mathcal{S} \times \mathbb{R}^k \to \mathbb{R}$ is a **shared observation function**—the somatotopic map. The dynamics on the manifold are phoneme-determined:

$$\dot{\mathbf{z}}(t) = h\big(\mathbf{z}(t);\ \mathbf{y}\big) \tag{5}$$

This formulation **idealizes away patient-specific effects**: g is shared and $\mathbf{z}(t)$ is universal. In reality, the observation depends on patient identity through both geometry (T_i) and physiology ($r^{(i)}$). The implemented model (Section 10) restores patient conditioning. Equation (4) is the motivating abstraction; the model does not enforce it literally.

3.2 The factorization is the key insight

The shared field $f(\mathbf{p}, t; \mathbf{y})$ factorizes as $g(\mathbf{p}, \mathbf{z}(t))$, separating space from time through a low-dimensional bottleneck. The model learns:

1. A spatial function g: how each brain position encodes articulatory features (somatotopic map).
2. A dynamical system h: how the articulatory state evolves during speech.

3.3 Articulatory manifold dimensionality

Contrast	**Phonemes distinguished**	**Dim.**
Place of articulation	labial (b,p,v) vs. velar (g,k)	∼2
Manner	stop vs. fricative vs. vowel	∼2
Voicing	voiced (b,g,v) vs. voiceless (k,p)	∼1
Vowel height	low (a,æ) vs. high (i,u)	∼1
Vowel backness	front (æ,i) vs. back (a,u)	∼1
Effective dimension		**∼5–8**

These dimensions are heuristic (the factors are correlated, not orthogonal) and do not by themselves establish identifiability—unknown transforms, noise, and patient residuals remain. However, the effective dimension is far below electrode count (63–201), suggesting the problem is **not obviously underdetermined under a low-dimensional latent hypothesis**.

3.4 Cross-patient transfer as manifold alignment

Each patient's array gives a different projection of the shared latent state. As a **toy linear approximation** (not a claim about the true relationship):

$$\mathbf{d}^{(i)}(t) \approx W^{(i)}\mathbf{z}(t) + \boldsymbol{\epsilon}^{(i)} \tag{6}$$

where $W^{(i)} \in \mathbb{R}^{N_i \times k}$ is determined by electrode positions. In practice, $W^{(i)}$ also absorbs patient-specific physiology, gain differences, pathology effects, and preprocessing artifacts. MNI coordinates provide the geometric relationship between projections a priori, rather than requiring post-hoc alignment (CCA, Procrustes, CORAL—all failed in our experiments).

4 Decomposing Cross-Patient Variation

4.1 The spatial transform T_i

$$T_i(\mathbf{p}) = A_i \cdot \mathbf{p} + \mathbf{b}_i + \delta_i(\mathbf{p}) \tag{7}$$

- $A_i \in \mathbb{R}^{3\times 3}$: rigid or affine (rotation, scaling). MNI normalization estimates this.
- $\mathbf{b}_i \in \mathbb{R}^3$: translation (15–25 mm array placement offset).
- $\delta_i(\mathbf{p})$: nonlinear residual—sulcal geometry, local folding, **tumor mass effect** (geometric mapping error in 6/12 patients).

4.2 Inter-subject correspondence uncertainty

Not a fixed constant. Sources: native-space localization (sub-mm), brain shift correction (variable), atlas warping (dominant). Net: several mm for healthy anatomy, more near tumors.

- At somatotopy scale (∼15–20 mm): reliable.
- At electrode spacing (∼2 mm): unreliable.
- In between (∼5–10 mm): partially informative.

4.3 Resolution mismatch implies dual spatial encoding

Scale	Resolution	Best encoded by	What it captures
Cross-patient	∼10–15 mm	MNI coordinates	Articulatory somatotopy
Intra-patient local	∼2 mm	Grid topology	Fine-grained spatial patterns

4.4 The patient-specific residual $r^{(i)}$

Captures: (1) fine-grained neural organization varying across individuals; (2) signal characteristics (impedance, contact, SNR); (3) functional plasticity (accent, articulatory habits); (4) pathology-specific effects (Parkinson's excitability changes, tumor-adjacent reorganization).

5 What Our Experiments Reveal

5.1 The evaluation recipe dominates at this data scale

Backbone already extracts most signal; the bottleneck is extracting predictions from the representation.

Caveat: All improvements measured on S14 grouped-by-token 5-fold CV; per-fold effects are large (±0.05), so improvements <2 pp may not replicate on other patients.

Component	Category	PER improvement
Label smoothing 0.1	Training	−4.1 pp
Mixup α=0.2	Training	−1.7 pp
Per-position heads + dropout	Training	−1.2 pp
Weighted k-NN (k=10, cosine)	Evaluation	−2.1 pp
TTA (16 augmented copies)	Evaluation	−1.4 pp
Articulatory + CE dual head	Evaluation	−1.1 pp

5.2 Multi-scale temporal structure is validated

Multi-scale temporal encoding (stride 3+5+10) was the single best architectural improvement (PER 0.766 → 0.750). The three scales correspond to fast transitions (∼15 ms), syllable dynamics (∼25 ms), and phoneme-level patterns (∼50 ms).

Caveat: Validated with the current BiGRU backbone. Optimal stride values may differ for the transformer temporal encoder proposed here; re-validation is needed (Section 14).

5.3 SSL features carry partial structure

All SSL methods produce near-chance PER (0.85–0.91) under basic evaluation. JEPA + LP-FT achieves 0.797. SSL features are non-trivial but far weaker than supervised features.

5.4 The ∼0.76 PER convergence on S14

55 experiments converge to PER 0.750–0.780. **Attributing this to any single bottleneck would be premature**:

1. **S14 is no longer held-out.** 55+ architectures evaluated ⇒ it functions as a validation set.
2. **Measurement noise.** ∼30 samples per fold ⇒ ±0.05 PER noise. Differences <2 pp are within noise.
3. **Target-data scarcity.** S14 has 153 trials. Stage 2 contributes only ∼1.6 pp.
4. **Spatial alignment may be one factor.** Not isolated as *the* bottleneck.

6 The Case for MNI Coordinates

6.1 Currently unused information

The current pipeline uses per-patient Conv2d on grid topology. Each Conv2d is independent. The model has **zero information** about cross-patient spatial relationships. The shared backbone implicitly assumes feature alignment across patients—anatomically false for independent per-patient encoders.

6.2 What coordinates may provide

Knowing $\mathbf{p}_j^{(i)}$ allows learning g as a shared function. Concretely: (1) identifying which articulatory zone each electrode samples; (2) sharing representations across patients at similar positions; (3) contextualizing variable coverage.

6.3 What coordinates do NOT guarantee

Coordinates help only if correspondence uncertainty is small enough relative to somatotopic scale, the architecture exploits them, and spatial alignment is actually binding. **Their value must be demonstrated empirically.**

Part II

Cross-Disciplinary Foundations

7 The Multi-View Reconstruction Analogy

Inspired by holistic performance capture [16], the cross-patient decoding problem maps structurally to **fitting a parametric 3D model to 2D observations from multiple viewpoints**.

7.1 The pose estimation mapping

3D Pose Estimation		**Neural Field Decoding**
3D body model (SMPL mesh)	$\leftrightarrow$	Shared neural field $g(\mathbf{p}, \mathbf{z})$ on cortex
Pose parameters θ	$\leftrightarrow$	Articulatory state $\mathbf{z}(t)$
Shape parameters β	$\leftrightarrow$	Patient embedding $e^{(i)}$
Camera extrinsics $[R \mid t]$	$\leftrightarrow$	MNI offset $\Delta^{(i)}$ (+ optional $\omega^{(i)}$)
Camera intrinsics K	$\leftrightarrow$	Per-patient gain/bias
2D keypoints x_j	$\leftrightarrow$	Electrode observations $d_j^{(i)}(t)$
Projection function $\pi(\cdot)$	$\leftrightarrow$	Observation function $g(\cdot)$
Reprojection loss $\|\pi(V) - x\|^2$	$\leftrightarrow$	**Reprojection loss** $\|g(\mathbf{p}+\Delta, \hat{\mathbf{z}}) - d\|^2$

7.2 What the analogy reveals

Why previous methods failed: CCA/Procrustes/CORAL align 2D feature statistics—works for similar viewpoints, fails for 15–25 mm offsets. Per-patient SpatialConv is a separate 3D model per camera—no sharing. Our approach is **structurally analogous to Structure from Motion**: jointly estimate shared structure and all camera parameters.

Why reprojection loss beats arbitrary SSL: JEPA/VICReg/masked reconstruction used physically meaningless targets. Reprojection is geometrically grounded: "if your latent estimate is correct, you should predict what every electrode observes." Self-supervised but geometrically constrained.

7.3 Where the analogy breaks

The mapping is structural, not exact. Three important disanalogies:

1. **Each patient is a different "scene," not a different camera on the same scene.** In multi-view reconstruction, all cameras observe the SAME 3D scene simultaneously. Our patients are observed independently, at different times, producing different spatiotemporal patterns due

to $r^{(i)}$. The shared structure f exists but is weaker than multi-view photometric consistency. The reprojection loss is therefore noisier—it must reconstruct through patient-specific residual, not just geometric projection.

2. **Reprojection is electrode-sparse, not pixel-dense.** A 256×256 image provides 65K reprojection constraints per view. Our 63–201 electrodes provide 100–1000× fewer constraints per frame. The reprojection signal per frame is thin; it aggregates value primarily across the temporal dimension ($T \approx 200$ frames/trial × 1315 trials).

3. **The "scene" changes every frame.** In static reconstruction, the scene is fixed across views. Here, $\mathbf{z}(t)$ changes continuously, so each frame is a different scene for reprojection. This is closer to **dynamic scene reconstruction**—but the temporal structure is phoneme-specific and short-lived (∼1 s trials), not a persistent evolving scene.

These disanalogies mean: the reprojection loss will be noisier and less constraining than its vision analog; geometric inductive biases (distance bias, bilinear factorization) must compensate; and the reprojection loss is a useful regularizer but **should not be expected to carry the architecture** the way photometric loss carries NeRF. The supervised classification loss remains primary.

Transfer criterion. We adopt 3D vision patterns only when the following structural property holds: the component solves a problem of sparse, geometrically-referenced observations of a shared underlying field, which is our problem regardless of the source domain. Where the analogy demands dense observations, persistent scenes, or exact photometric consistency, we either adapt or drop the pattern.

8 Transferable Patterns from 3D Vision (2024–2026)

8.1 The field's trajectory: per-scene → feed-forward → foundation model

Our architecture sits at step 2 (feed-forward). Key transferable patterns:

VGGT [16]: joint 3D attribute prediction from variable-length views. Validates Perceiver jointly estimating latent state + alignment.

LVSM [17]: encoder-decoder with fixed latent tokens vs. decoder-only. Decoder-only wins with enough data; explicit structure wins at small scale. *Validates our structured approach for* ∼1300 *trials.*

MMGN [18]: continuous field reconstruction from sparse, irregularly sampled sensors using factorized spatiotemporal basis. Our bilinear decoder is structurally related (Section 13).

SPARF [19]: NeRF from sparse and noisy poses. Directly analogous: our MNI coordinates are noisy poses; our electrodes are sparse views.

Senseiver [20]: Perceiver IO for field reconstruction from sparse sensors—structurally related to our spatial encoder.

8.2 Architectural pattern validation

8.3 What we are correctly NOT doing

Diffusion priors (need millions of scenes), 4D Gaussian primitives (trials are independent events), foundation model scale (1300 trials may struggle to support >200K params—our budget is at the

Our component	Same pattern in vision
Perceiver cross-attention (virtual → actual)	Canonical-to-observation tokens (LRM, LVSM)
Virtual electrodes at learnable positions	Triplane NeRF / fixed latent tokens
MNI distance-biased attention	Geometry-aware attention (GraphSplat, MVSplat)
Bilinear decoder $\hat{d} = \phi(\mathbf{p})^\top \psi(s)$	Spatiotemporal factorization (MMGN)
Reprojection loss	Photometric loss (NeRF/3DGS family)
Learnable per-patient Δ	Learnable camera extrinsics (SPARF, AnySplat)
Patient embedding $e^{(i)}$	Per-scene latent code (CodeNeRF)
Electrode masking	View dropout / masked pretraining (MAE)
Structured inductive bias for small data	Explicit geometry at small scale (LVSM)

edge), decoder-only architecture (needs massive data), contrastive SSL (failed in our experiments).

8.4 The data scale gap

	3D Vision	Our Problem
Training scenes/patients	50K–1M+	∼20 patients
Observations per scene	10–100+ images (M pixels)	63–201 electrodes
Total data	TB scale	∼240 MB
Parameter budget	50M–500M+	∼200K

This gap means: we **must** use strong geometric inductive biases; the reprojection loss is valuable but electrode-sparse (Section 7.3); and our ∼200K architecture is appropriately small. The field's finding that explicit geometric structure helps at small scale directly supports this design.

9 Uncertainty Estimation

9.1 From binary to continuous confidence

The current pipeline has a binary mechanism: significant channel selection (keep/drop). Proper uncertainty replaces this with continuous per-electrode confidence.

9.2 Two types of uncertainty

Aleatoric (irreducible): "This electrode position tends to have low SNR." Per-electrode, does not decrease with more data.

Epistemic (reducible): "We lack data in this cortical region." Per-position, decreases with more patients.

9.3 Recommended uncertainty stack

9.3.1 Tier 1: Minimal implementation, high value

Beta-NLL loss [13]. Replace MSE reprojection with learned per-electrode variance:

$$\mathcal{L}_{\text{reproj}}^{\text{UQ}} = \frac{1}{2}\Big(\log \hat{\sigma}_j^2 + \frac{(d_j - \hat{d}_j)^2}{\hat{\sigma}_j^2}\Big) \cdot \big(\hat{\sigma}_j^2\big)_{\text{sg}}^{0.5} \tag{8}$$

Scope: Beta-NLL affects the **training loss only**—it downweights reprojection gradients from noisy electrodes, preventing them from distorting the encoder. It does **not** directly modulate attention weights at inference time.

Static electrode reliability gate (for inference-time uncertainty weighting). To propagate position-dependent reliability into the Perceiver attention, we add a per-electrode reliability scalar that depends only on electrode identity—not on current neural activity:

$$c_j^{\text{rel}} = \sigma\Big(\text{Linear}\big(\text{MNI_PE}(\tilde{\mathbf{p}}_j) + \text{Grid_PE}(\text{row}_j, \text{col}_j)\big)\Big) \in (0, 1) \tag{9}$$

The reliability gate scales attention **values** (not logits): $V_j \leftarrow c_j^{\text{rel}} \cdot V_j$ before aggregation. Because c_j^{rel} depends only on positional features, it can be legitimately interpreted as "this electrode position tends to be unreliable"—capturing effects like edge-of-array contact quality, proximity to dead electrodes, or cortical regions with low speech-related activity. It adds $d+1 = 65$ trainable parameters.

Assumption: We deliberately exclude current activity from the reliability gate to preserve interpretability. A dynamic content gate $c_j^{\text{dyn}} = \sigma(\text{Linear}(\text{token}_j))$ could learn richer routing but can no longer be interpreted as electrode quality—it may suppress rare but real speech-related activity. If needed, a dynamic gate can be added as a separate mechanism (ablation A7b), but the default is position-only.

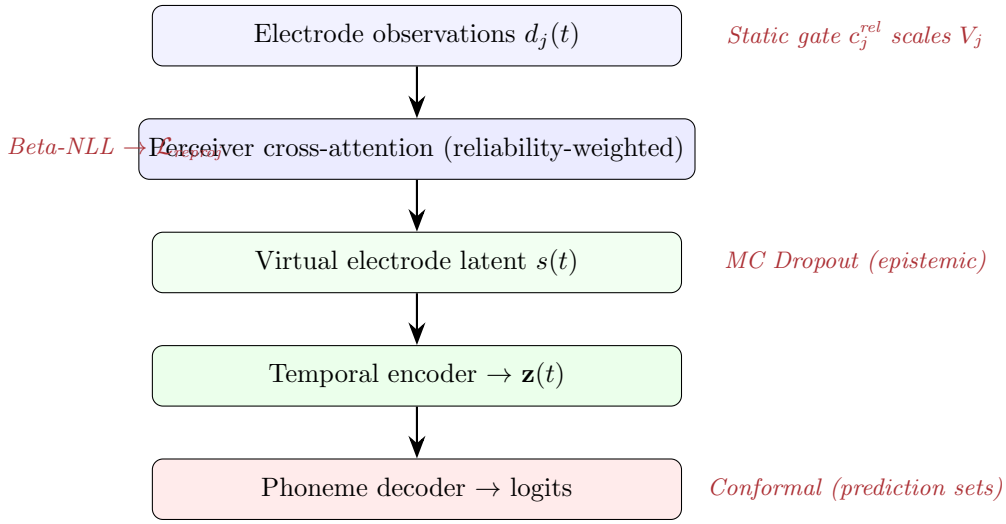
MC Dropout at inference [14]. Keep dropout active, 10–20 passes, variance = epistemic uncertainty. Remove dropout from output layer.

Conformal prediction [15]. Post-hoc prediction **sets** with coverage guarantees. Use CV fold as calibration. Potential interpretability value: "model thinks /b/ or /p/ (both labial stops)"—the confusion structure itself is informative.

9.3.2 Tier 2: Moderate implementation

Evidential regression (NIG prior): single-pass decomposed aleatoric + epistemic per electrode. **Probabilistic virtual electrodes** (pixelSplat-style): predict latent distributions, uncertainty propagates through the pipeline.

9.4 How uncertainty connects to the architecture



Beta-NLL affects training gradients (left). Static reliability gate c_j^{rel} affects inference-time attention (right). MC Dropout captures model uncertainty. Conformal prediction gives calibrated prediction sets. Each mechanism has a distinct scope—they are complementary, not redundant.

Part III

Architecture Design

10 The Implemented Model

The idealized equations in Part I motivate the architecture but are not what the model computes. The implemented model is:

$$\boxed{\hat{d}_j^{(i)}(t) = g_\theta\big(\tilde{\mathbf{p}}_j^{(i)},\ s^{(i)}(t),\ e^{(i)}\big) + \epsilon_j^{(i)}(t)} \tag{10}$$

where:

- $\tilde{\mathbf{p}}_j^{(i)} = R(\omega^{(i)}) \cdot \mathbf{p}_j^{(i)} + \Delta^{(i)}$ is the geometrically corrected electrode position.
- $s^{(i)}(t) \in \mathbb{R}^{L \times d}$ is the spatial encoder output for patient i at time t (depends on all of patient i's electrode observations and patient embedding).
- $e^{(i)} \in \mathbb{R}^d$ is the patient identity embedding.
- g_θ is the observation decoder (Section 13), parameterized by shared weights θ.

How this relates to the idealizations.

- Equation (4) posits a *shared* observation function $g(\mathbf{p}, \mathbf{z})$ with no patient dependence. The implemented model conditions on patient identity through both $e^{(i)}$ (in the decoder) and the corrected positions $\tilde{\mathbf{p}}_j^{(i)}$ (patient-specific geometric correction). This is a deliberate relaxation: it allows the model to handle patient-specific residuals $r^{(i)}$ that the idealized equation assumes away.
- The "shared" component lives in the shared weights θ and the shared virtual electrode positions. The patient-specific components $(\Delta^{(i)}, \omega^{(i)}, e^{(i)})$ are small relative to θ ($\sim$70 vs. $\sim$200K parameters), so the model is biased toward sharing but not constrained to it.
- The decoder uses $s(t)$ (per-frame spatial estimate), not $\mathbf{z}(t)$ (temporally-refined state). This is because reprojection is a per-frame spatial task. The temporal encoder produces $\mathbf{z}(t)$ for classification only.

11 Design Principles

Every component maps to the observation function, with an explicit distinction between two latent representations:

The $s(t)$ vs. $\mathbf{z}(t)$ distinction. The theory posits a single latent state $\mathbf{z}(t)$. The architecture produces two representations at different processing stages:

Symbol	Meaning	Architectural component
$d_j^{(i)}(t)$	Electrode observation	Input token
$\mathbf{p}_j^{(i)}$	MNI position (approximate)	MNI positional encoding
$\Delta^{(i)}, \omega^{(i)}$	Registration correction (translation + rotation)	Learnable 6D offset
$s(t) \in \mathbb{R}^{L\times d}$	**Per-frame spatial summary**	Spatial encoder output
$\mathbf{z}(t) \in \mathbb{R}^d$	**Temporally-refined state**	Temporal encoder output
$e^{(i)}$	Patient residual	Learnable embedding
$g_\theta(\cdot)$	Observation decoder (patient-conditioned)	Bilinear decoder

- $s(t) \in \mathbb{R}^{L\times d}$: the spatial encoder output. Per-frame, no temporal context. Captures the instantaneous spatial pattern of cortical activity across L virtual electrodes.
- $\mathbf{z}(t) \in \mathbb{R}^d$: the temporal encoder output. Incorporates multi-scale temporal context from neighboring frames.

The observation decoder uses $s(t)$ for reprojection because it reconstructs a per-frame spatial pattern. The phoneme decoder uses $\mathbf{z}(t)$ because phoneme classification requires temporal dynamics. $s(t)$ is **not** the articulatory state $\mathbf{z}$ from Eq. (4)—it is the architecture's per-frame estimate of the spatial field, from which $\mathbf{z}(t)$ is later refined.

Objective tension. $\mathcal{L}_{\text{reproj}}$ wants $s(t)$ to preserve all variability (for faithful reconstruction); $\mathcal{L}_{\text{class}}$ wants $s(t)$ to be discriminative (discarding irrelevant variance). These objectives share the spatial encoder and can compete. The temporal encoder provides a buffer, and λ annealing ($0.5 \to 0.1$) gradually prioritizes classification. If the tension is severe, λ=0 (pure supervised) is the fallback (tested in ablation A5).

12 Spatial Encoder (∼100K params)

12.1 Electrode tokens

For electrode j on patient i at time t:

$$\text{token}_j = \underbrace{\text{Linear}(d_j \to d)}_{\text{activity}} + \underbrace{\text{Linear}(\tilde{\mathbf{p}}_j \to d)}_{\text{MNI PE}} + \underbrace{\text{Linear}(\text{row}_j, \text{col}_j \to d)}_{\text{Grid PE}} + \underbrace{e^{(i)}}_{\text{patient}} \tag{11}$$

where $\tilde{\mathbf{p}}_j = R(\omega^{(i)}) \cdot \mathbf{p}_j + \Delta^{(i)}$ is the corrected MNI position.

Per-patient geometric parameters ($6 + d$ per patient):

- $\Delta^{(i)} \in \mathbb{R}^3$: learnable translation, initialized to $\mathbf{0}$.
- $\omega^{(i)} \in \mathbb{R}^3$: learnable rotation (axis-angle $\to$ rotation matrix via Rodrigues), initialized to $\mathbf{0}$. Corrects array rotation/orientation error.
- $e^{(i)} \in \mathbb{R}^d$: learnable patient embedding, initialized $\sim \mathcal{N}(0, 0.02)$.

Assumption: The patient embedding $e^{(i)}$ is added uniformly to all electrode tokens. This means the encoder cannot apply **position-dependent** patient-specific corrections—only the observation decoder can (via $\text{MLP}_{\text{spatial}}(\tilde{\mathbf{p}}, e)$). For tumor patients with spatially non-uniform geometric distortion $\delta_i(\mathbf{p})$, this is a known limitation. The rotation $\omega^{(i)}$ partially addresses this (correcting rigid misalignment beyond translation), but cannot capture nonlinear warping. Ablation A3 tests whether richer per-patient parameterization (e.g., position-dependent $e^{(i)}$) helps.

Dead electrodes are excluded (not masked). Variable N_i is handled natively by attention.

12.2 Perceiver cross-attention with distance bias

$L = 16$ **virtual electrodes** with learnable MNI positions, initialized to cover the convex hull of source electrode positions:

$$\text{query}_i = \text{embed}_i + \text{MNI_PE}(\text{v_pos}_i) \tag{12}$$

Cross-attention with spatial locality prior:

$$\boxed{A_{ij} = \frac{Q_i \cdot K_j}{\sqrt{d}} \; - \; \alpha \cdot \|\text{v_pos}_i - \tilde{\mathbf{p}}_j\|^2} \tag{13}$$

where $\alpha > 0$ is learnable (initialized 0.1), controlling spatial blur.

Attention values are scaled by the static reliability gate (Section 9):

$$\text{out}_i = \sum_j \text{softmax}(A)_{ij} \cdot \left(c_j^{\text{rel}} \cdot V_j\right) \tag{14}$$

where $c_j^{\text{rel}} = \sigma(\text{Linear}(\text{MNI_PE}(\tilde{\mathbf{p}}_j) + \text{Grid_PE}(\text{row}_j, \text{col}_j)))$. Electrodes at unreliable positions contribute less.

4-head attention, optional 1-layer self-attention (captures inter-zone co-articulation).

12.3 Spatial readout: attention pooling (not MaxPool)

The $L = 16$ virtual electrode outputs form a **spatially-structured representation** (L, d) where each virtual electrode captures activity from a different cortical zone. Naïve MaxPool would discard which zone contributed what—destroying the spatial decomposition the virtual electrodes were designed to provide.

Instead, we use **attention-based readout**:

$$s_{\text{pool}}(t) = \sum_{i=1}^{L} w_i \cdot \text{out}_i(t), \qquad w_i = \text{softmax}\left(\mathbf{q}_{\text{read}}^{\top} \text{out}_i(t)/\sqrt{d}\right) \tag{15}$$

where $\mathbf{q}_{\text{read}} \in \mathbb{R}^d$ is a single learnable readout query. This is a soft selection over zones—preserving the somatotopic decomposition through learned weighting.

Full spatial output: $s(t) \in \mathbb{R}^{L \times d}$ (all virtual electrode outputs) is preserved for the observation decoder. $s_{\text{pool}}(t) \in \mathbb{R}^d$ is the default input to the temporal encoder.

12.4 Parameter count

13 Observation Decoder (∼25K params)

The forward observation model—the "renderer" in the multi-view analogy. Uses the full spatial output $s(t) \in \mathbb{R}^{L \times d}$.

Component	Params
Activity embedding, MNI PE, Grid PE	576
Virtual embeddings + positions	1,072
Cross-attention + self-attention + FFN	∼98K
Reliability gate	65
Readout query	64
Distance scale α	1
Total spatial encoder	**∼100K**
Per-patient $(\Delta + \omega + e)$	$6 + d$/patient

13.1 Position-weighted latent aggregation

To reconstruct what electrode j observes, the decoder needs a latent summary that is **local to electrode j's position**. Using $\bar{s}(t) = \text{mean}(s(t))$ would destroy the spatial structure that the virtual electrodes provide—nearby and distant virtual electrodes would contribute equally, regardless of cortical position.

Instead, each electrode j aggregates virtual electrode outputs weighted by spatial proximity:

$$s_j^*(t) = \sum_{\ell=1}^{L} a_{j\ell} \cdot \text{out}_\ell(t) \in \mathbb{R}^d, \qquad a_{j\ell} = \frac{\exp\left(-\alpha_{\text{dec}} \cdot \|\tilde{\mathbf{p}}_j - \text{v_pos}_\ell\|^2\right)}{\sum_{m=1}^{L} \exp\left(-\alpha_{\text{dec}} \cdot \|\tilde{\mathbf{p}}_j - \text{v_pos}_m\|^2\right)} \tag{16}$$

where $\alpha_{\text{dec}} > 0$ is a learnable spatial scale (initialized to match the encoder's α). Virtual electrodes near real electrode j contribute more; those covering distant cortical zones contribute less. This is parameter-free except for α_{dec} (1 parameter) and reuses the virtual electrode positions already learned by the spatial encoder.

13.2 Bilinear observation model

$$\boxed{\hat{d}_j^{(i)}(t) = \phi(\tilde{\mathbf{p}}_j,\ e^{(i)})^\top\ \psi\big(s_j^*(t)\big)} \tag{17}$$

where:

- $\phi = \text{MLP}_{\text{spatial}}(\tilde{\mathbf{p}}, e) \in \mathbb{R}^k$: spatial tuning vector at corrected position.
- $\psi = \text{MLP}_{\text{latent}}(s_j^*) \in \mathbb{R}^k$: articulatory activation from the position-local latent.
- $k = 32$: bilinear interaction dimensionality.

Note: The decoder takes $s_j^*(t)$, not $\mathbf{z}(t)$. Since reprojection is a per-frame spatial task (reconstructing electrode observations from the instantaneous field estimate), temporal context is unnecessary. The temporal encoder is trained only through $\mathcal{L}_{\text{class}}$.

Assumption: The bilinear form $\hat{d} = \phi(\mathbf{p})^\top \psi(s)$ assumes the observation function **separates** into spatial tuning and articulatory activation interacting through an inner product. This means: the response of an electrode at position $\mathbf{p}$ to state s is a linear combination of k independent modes. Higher-order interactions (e.g., coarticulation effects where tongue position modulates lip-area neural activity) cannot be captured. This is a strong structural assumption, validated by MMGN [18] for physical fields but unverified for coarticulatory neural dynamics. Ablation A9 tests a full MLP fallback $\hat{d} = \text{MLP}([\tilde{\mathbf{p}}; e; s_j^*])$ at the cost of losing the factorization's interpretability and inductive bias.

Position-dependent patient correction. Unlike the encoder (where $e^{(i)}$ is a global additive bias), the decoder's $\mathrm{MLP}_{\mathrm{spatial}}(\tilde{\mathbf{p}}, e)$ can learn position-dependent patient effects: "for this patient, electrodes near position $\mathbf{p}$ respond differently." This provides the capacity to partially model the nonlinear residual $\delta_i(\mathbf{p})$ that the global $\Delta^{(i)}, \omega^{(i)}$ cannot.

13.3 Reprojection loss (self-supervised)

$$\mathcal{L}_{\mathrm{reproj}} = \frac{1}{N_i \cdot T} \sum_{t=1}^{T} \sum_{j=1}^{N_i} \left\| d_j(t) - \hat{d}_j(t) \right\|^2 \tag{18}$$

Electrode masking: Drop 20–40% of electrodes from encoder input; keep them in reprojection loss. The model must spatially interpolate—geometric masked reconstruction.

14 Temporal Encoder (∼70K params)

14.1 Multi-scale tokenization

$$\begin{aligned} s_{\mathrm{pool}}(1{:}T) &\xrightarrow{\mathrm{Conv1d}(d,d,k=3,s=3)} T_3 \approx 67 \text{ tokens at } \sim 67\,\mathrm{Hz} \\ s_{\mathrm{pool}}(1{:}T) &\xrightarrow{\mathrm{Conv1d}(d,d,k=5,s=5)} T_5 \approx 40 \text{ tokens at } \sim 40\,\mathrm{Hz} \\ s_{\mathrm{pool}}(1{:}T) &\xrightarrow{\mathrm{Conv1d}(d,d,k=10,s=10)} T_{10} \approx 20 \text{ tokens at } \sim 20\,\mathrm{Hz} \end{aligned} \tag{19}$$

All ≈127 tokens concatenated with learnable per-scale embeddings + sinusoidal temporal PE.

Caveat: Multi-scale stride 3+5+10 was validated on BiGRU. The temporal transformer processes sequences differently (parallel attention vs. recurrent). The articulatory timescales are physics-based (not architecture-specific), so the stride values should be approximately correct, but the optimal configuration should be re-validated (ablation A10).

14.2 Temporal transformer

2 layers, 4 heads, d=64, FFN dim 256. Pre-norm transformer. Each token attends across all scales and times.

14.3 Readout

Global readout: $z_{\mathrm{global}} = \mathrm{mean}(\text{all tokens}) \in \mathbb{R}^d$ for k-NN.
Per-position: $z_p = \mathrm{mean}(\text{tokens in position window}) \in \mathbb{R}^d$ for classification.

15 Phoneme Decoder (∼5K params)

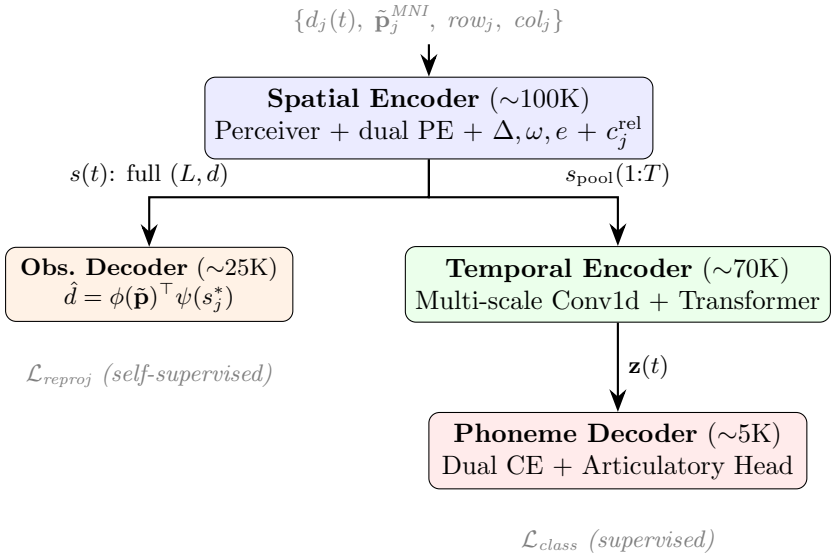
Dual head:

$$\mathrm{logits}_p = \frac{1}{2}\Big(\underbrace{\mathrm{Linear}(\mathbf{z} \to 9)}_{\text{CE head}} + \underbrace{\tau \cdot \mathrm{normalize}(\mathrm{Linear}(\mathbf{z} \to 15)) \cdot A^\top}_{\text{articulatory head}}\Big) \tag{20}$$

where $A \in \mathbb{R}^{9\times 15}$ is the fixed articulatory feature matrix and τ is learnable temperature.

$\mathcal{L}_{\mathrm{class}} = \frac{1}{3}\sum_{p=1}^{3} \mathrm{FocalCE}(\mathrm{logits}_p, y_p;\ \gamma{=}2, \mathrm{LS}{=}0.1)$ with mixup (α=0.2).

16 Complete Architecture



$$\boxed{\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda \cdot \mathcal{L}_{\text{reproj}}} \tag{21}$$

Module	**Params**	**Role**
Spatial encoder	∼100K	Invert observation function
Observation decoder	∼25K	Forward model (reprojection)
Temporal encoder	∼70K	Dynamics modeling
Phoneme decoder	∼5K	Classification
Total shared	**∼200K**	
Per-patient (×10)	700	Registration + identity
Grand total	**∼201K**	(cf. current pipeline: 122K)

Part IV

Training, Evaluation, and Experimental Design

17 Training Pipeline

17.1 Phase 1: Source patient training (supervised + reprojection)

9 source patients, ∼1315 labeled trials. Each batch:

1. **Spatial encoding**: construct tokens, mask 20–40% electrodes, Perceiver → $s(t)$, attention pooling → $s_{\text{pool}}(t)$.
2. **Reprojection**: decoder predicts all electrodes (including masked) from position-local latents $s_j^*(t)$; compute $\mathcal{L}_{\text{reproj}}$ (or $\mathcal{L}_{\text{reproj}}^{\text{UQ}}$).

3. **Temporal encoding**: $s_{\text{pool}}(1{:}T) \to$ multi-scale $\to$ transformer $\to \mathbf{z}(t)$.
4. **Classification**: dual head $\to \mathcal{L}_{\text{class}}$.
5. **Total**: $\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda \mathcal{L}_{\text{reproj}}$ (λ: $0.5 \to 0.1$ annealed).

Optimizer: AdamW with differential LR:

- Shared parameters: 10^{-3}
- Virtual positions, Δ, ω: 3×10^{-4} (slow—geometric stability)
- Patient e: 10^{-3}

Linear warmup (20 epochs) $\to$ cosine decay. Weight decay 10^{-4}. Gradient clipping 5.0. Batch size 16. Early stopping on held-out 20% source validation, patience 7.

17.2 Phase 2: Target adaptation (LP-FT)

Per CV fold on target patient (5-fold grouped-by-token):

1. **Linear probe**: freeze encoder, train decoder + $\Delta^{(\text{target})} + \omega^{(\text{target})} + e^{(\text{target})}$. Initialize $e = \text{mean}(e^{(\text{source})})$. 150 epochs. Loss: $\mathcal{L}_{\text{class}} + 0.1 \cdot \mathcal{L}_{\text{reproj}}$. **Best checkpoint selected on inner validation split** (20% of training fold held out).
2. **Fine-tune**: unfreeze encoder at 0.1× LR. 100 epochs, patience 7. Best checkpoint by inner validation.

17.3 Phase 3 (optional): Transductive reprojection pre-training

Pre-train spatial encoder + observation decoder using $\mathcal{L}_{\text{reproj}}$ only.

Protocol distinction. This phase uses unlabeled neural data for self-supervised pre-training. If the target patient's unlabeled data is included, this is a **transductive** protocol—the model sees target recordings before supervised evaluation. Two regimes must be reported separately:

1. **Strict LOPO**: pre-training uses only source patient data. Target patient is completely held out until Phase 2.
2. **Transductive SSL**: pre-training includes target patient's unlabeled recordings (no labels, but the model sees the distribution). This is legitimate but **not comparable to strict LOPO** without explicit disclosure.

Phase 1 is then initialized from pre-trained weights.

18 Evaluation Protocol

18.1 Three evaluation regimes

The cross-patient setting admits three regimes of increasing difficulty. The scientific claim differs across regimes, and results should never be compared across regimes without explicit acknowledgment:

1. **Zero-shot transfer.** Train on sources (Phase 1). Evaluate on target with no adaptation. Claim: "the shared model generalizes to unseen electrode configurations." This is the strongest scientific claim but likely yields poor PER at our data scale.

2. **LP-FT adaptation** (primary evaluation). Phase 1 → Phase 2 (linear probe + fine-tune on target folds). Claim: "source training provides a useful initialization for few-shot target adaptation." This is our primary regime—it matches the clinical scenario where a short calibration session is available.

3. **Transductive SSL + LP-FT.** Phase 3 (with target unlabeled data) → Phase 1 → Phase 2. Claim: "unlabeled target data further improves adaptation." Reported separately from regime 2.

18.2 Metrics and protocol

PER via edit distance, grouped-by-token 5-fold CV. Full recipe: weighted k-NN (k=10, cosine), TTA 16, dual head, source k-NN at 0.5× weight. Conformal prediction sets as interpretability output.

Checkpoint selection. Within each CV fold, the best checkpoint is selected on a **held-out inner validation split** (20% of the training portion of that fold). The test portion of the fold is never used for model selection. This is standard nested cross-validation, not test-set leakage.

19 Ablation Experiments

ID	Ablation	Question
A1	MNI PE present/absent	**Central**: do MNI coordinates help?
	a. Current Conv2d + BiGRU baseline	→ 0.762 (LP-FT regime)
	b. Perceiver + Grid PE only	→ architecture benefit
	c. Perceiver + MNI + Grid PE	→ MNI benefit
	d. (c) + reprojection loss	→ reprojection benefit
A2	**Population-level (all patients)**	**Is the 0.76 wall S14-specific?**
A3	Per-patient params	Δ only / $+\omega$ / $+e$ / position-dep. e
A4	Virtual electrode count	$L \in \{4, 8, 16, 32\}$
A5	Reprojection weight	$\lambda \in \{0, 0.01, 0.1, 0.5\}$
A6	Cross-task pooling	+ Duraivel data (architecture-independent)
A7	Uncertainty stack	a. Beta-NLL + static gate vs. MSE-only
		b. Static gate vs. dynamic gate
A8	Spatial readout	MaxPool / attention pool / full (L, d)
A9	Bilinear vs. MLP decoder	$\phi^\top \psi$ vs. MLP($[\tilde{\mathbf{p}}; e; s_j^*]$)
A10	Temporal backbone	Transformer vs. BiGRU (+ stride re-valid.)

20 Experimental Priorities

The document's own evidence suggests a clear priority ordering that should be stated transparently:

1. **Cross-task data pooling** (A6). Lowest risk, highest confidence. Same phonemes, overlapping patients, validated by Duraivel 2025 [12]. Directly addresses target-data scarcity. Can be tested with the current pipeline **and** the new architecture. Should be pursued first, in parallel with coordinate sourcing.

2. **Population-level evaluation** (A2). Pure methodology fix. Required to determine whether the 0.76 wall is S14-specific (55-experiment overfitting) or universal. Must be the **primary success criterion**, not just an ablation. Zero architectural risk.

3. **MNI coordinate-based architecture** (A1, this document). A working hypothesis. The evidence that spatial alignment is THE binding constraint is suggestive but unvalidated: it has not been isolated from data scarcity, measurement noise, or S14-specific effects (Section 5.4). The architectural investment is justified by: (a) MNI coordinates are currently unused information; (b) the observation function formulation provides a principled framework; (c) every component has a validated analog in related work. But the hypothesis could be wrong—coordinates may not help if the wall is data-limited rather than alignment-limited.

The architecture is the most speculative of the three paths. This is acceptable—it is the path with the most intellectual contribution if it works—but the document should not obscure the priority ordering.

21 Success Criteria

Primary criterion (population-level):

- Mean PER across all patients improves over current LOPO baseline with best architecture (A1c/d vs. A1a), evaluated across $\geq$3 seeds, in the LP-FT regime.
- This is the only criterion that provides scientific evidence for the MNI coordinate hypothesis.

S14 pilot evidence (exploratory only):

- S14 results serve as **pilot directional evidence** during development. They are contaminated by 55+ prior experiments and should not be used as definitive success criteria.
- If population-level results disagree with S14 pilot results, the population-level results take precedence.

Interpretability (qualitative validation):

- Virtual electrode positions cluster over known speech motor regions.
- Learned $\Delta^{(i)}, \omega^{(i)}$ are weak proxies for registration error unless independently validated (e.g., against known registration quality metrics). They may also absorb physiology, coverage differences, or optimization artifacts. Correlation with tumor status would be suggestive but not conclusive.
- Spatial tuning vectors $\phi(\mathbf{p})$ recover articulatory somatotopy when visualized.
- Reliability gate c_j^{rel} correlates with electrode impedance or SNR—interpretable because it depends only on position, not current activity.

22 Available Data

Data source	Volume	Usage	Risk
PS task HGA	9 pts, ∼1315 trials	Source training	–
S14 HGA	153 trials	Target eval (pilot)	–
Cross-task (Duraivel)	3 pts, 156–208 trials	**Priority 1: more data**	Low
MNI coordinates	All patients	Perceiver architecture	Medium
Continuous recordings	∼168 min	Phase 3 reproj. SSL	Medium
Lexical task patients	13 patients	Additional sources	Medium

23 MNI Coordinate Sourcing

Hard dependency. Likely locations: BioImage Suite outputs (DBS), Brainlab (tumor), ECoG_Recon on Box. Fallback: scaled ACPC grid positions (retains within-patient structure, loses cross-patient alignment).

24 Risk Assessment

Risk	**Sev.**	**Mitigation**
MNI coordinates unavailable	High	Fallback to Grid PE only
MNI alignment too imprecise	High	Ablation A1; learnable α adapts blur
Reprojection competes with classification	Medium	λ ablation A5; λ=0 fallback
Perceiver overfits (200K, 1300 trials)	Medium	Masking; early stopping; reprojection reg.
Bilinear separability too restrictive	Medium	Ablation A9: MLP fallback
Per-patient params insufficient for tumor	Medium	Ablation A3; position-dep. e option
Virtual electrodes collapse	Low	Initialize across convex hull; monitor

25 Summary

This document formalizes cross-patient speech decoding as **multi-view neural field reconstruction**, a structural analogy to 3D vision that is generative but not a substitute for empirical validation (Section 7.3).

The implemented model (Eq. 10) is honestly patient-conditioned: it conditions on patient identity $e^{(i)}$ and learned geometric corrections $\Delta^{(i)}, \omega^{(i)}$, relaxing the idealized shared observation function. The architecture produces two distinct representations: $s(t)$ (per-frame spatial summary, trained via reprojection) and $\mathbf{z}(t)$ (temporally-refined state, trained via classification). The observation decoder uses position-weighted aggregation of virtual electrode outputs (Eq. 16), preserving the spatial decomposition that motivates virtual electrodes. It employs a bilinear factorization that assumes separability between spatial tuning and articulatory activation—a structural hypothesis requiring empirical validation (ablation A9).

Key improvements over v3: (1) implemented model stated as the primary equation, with idealizations clearly labeled; (2) position-weighted latent aggregation in the decoder resolving the mean-pooling contradiction; (3) static reliability gate (position-only) separated from dynamic content gating; (4) three evaluation regimes defined (zero-shot, LP-FT, transductive); (5) Phase 3 explicitly flagged as transductive; (6) checkpoint selection clarified as inner-validation, not test-set leakage; (7) S14 demoted to exploratory pilot evidence; (8) overclaims reduced throughout; (9) bibliography added; (10) manifold "well-posed" claim softened; (11) "27-way classification" corrected to three 9-way; (12) linearized projection model marked as toy abstraction; (13) geometric correction interpretation caveated; (14) transfer criterion for 3D vision patterns made explicit.

Total budget: ∼200K shared + 70 per patient. The architecture is the most speculative of three research paths (Section 20); its value lies in principled exploitation of currently unused coordinate information, with every structural assumption marked for empirical test.

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